

VITORIA AP, Spain

WMO: 080800

Lat: 42.872N

Lon: 2.733W

Elev: 513

StdP: 95.32

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-2.5	-5.9	2.4	-1.9	-4.1	2.8	-0.5	12.2	10.8	10.5	9.7	0.8	20	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	33.5	20.2	31.0	19.7	28.8	19.1	21.6	29.9	20.7	28.6	19.7	26.8	3.5	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.8	24.3	18.1	13.9	23.0	17.3	13.1	22.0	65.2	30.0	61.7	28.4	58.5	27.0	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.1	7.1	-7.8	37.5	2.7	1.8	-9.8	38.8	-11.3	39.9	-12.8	40.9	-14.7	42.3		
			-8.0	23.7	2.7	1.1	-10.0	24.5	-11.6	25.1	-13.1	25.7	-15.1	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	5.6	5.9	8.7	10.5	13.8	17.6	19.6	20.0	17.2	13.7	8.5	5.7
	DBStd	6.22	3.26	3.27	3.39	3.23	3.51	3.69	3.17	3.34	3.17	3.42	3.23	3.37
	HDD10.0	574	142	119	65	32	7	0	0	0	0	8	66	137
	HDD18.3	2412	396	348	299	234	146	58	22	19	58	146	294	391
	CDD10.0	1409	4	4	24	47	126	228	299	311	217	124	21	4
	CDD18.3	204	0	0	0	0	6	36	63	72	24	4	0	0
	CDH23.3	2447	0	0	3	21	140	482	666	769	318	47	0	0
	CDH26.7	966	0	0	0	2	37	194	272	353	103	4	0	0
Wind	WSAvg	2.6	2.7	2.8	2.8	2.7	2.5	2.5	2.7	2.5	2.1	2.2	2.6	2.6
Precipitation	PrecAvg	995	107	101	98	95	81	58	35	35	58	87	136	105
	PrecMax	1498	278	285	234	200	220	147	98	84	119	325	262	201
	PrecMin	764	6	13	16	26	26	15	3	10	22	11	44	15
	PrecStd	162	64	71	59	44	41	33	24	21	27	60	51	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.6	18.8	23.0	26.0	30.5	34.9	35.6	36.9	33.1	27.1	19.2	15.7
		MCWB	11.8	11.0	13.3	15.5	18.3	20.7	20.6	20.7	19.4	16.8	14.0	11.8
	2%	DB	13.7	15.5	20.2	22.9	26.9	31.3	32.4	33.8	29.5	24.6	17.1	13.8
		MCWB	10.5	10.2	12.2	14.1	17.3	19.7	20.1	20.2	18.4	16.2	12.7	10.7
	5%	DB	12.0	13.4	17.9	20.0	24.2	28.8	29.8	30.8	27.0	22.1	15.2	12.1
		MCWB	9.3	9.3	11.3	13.1	16.0	19.2	19.7	19.9	18.0	15.5	11.8	9.5
	10%	DB	10.8	11.8	15.5	17.4	21.6	25.8	27.1	27.8	24.7	20.1	13.8	11.0
		MCWB	8.3	8.5	10.4	11.7	15.0	18.0	19.3	19.3	17.2	14.9	10.9	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	12.1	14.4	16.4	19.6	22.0	22.8	23.0	21.0	18.4	15.2	12.2
		MCDB	14.7	16.3	20.7	24.4	27.9	31.1	30.1	31.9	29.4	24.3	17.9	14.8
	2%	WB	10.8	10.7	12.9	14.7	17.9	20.7	21.6	21.6	19.6	17.2	13.3	10.9
		MCDB	12.9	14.6	18.5	21.4	25.7	29.2	29.3	30.2	26.9	22.4	16.3	13.1
	5%	WB	9.6	9.7	11.7	13.4	16.6	19.6	20.5	20.5	18.6	16.2	12.3	9.9
		MCDB	11.6	12.9	16.7	18.9	23.1	27.2	27.8	28.5	25.0	20.6	14.8	11.9
	10%	WB	8.6	8.7	10.6	12.2	15.4	18.5	19.5	19.6	17.7	15.3	11.2	9.0
		MCDB	10.3	11.2	15.0	16.7	20.7	24.9	25.9	26.2	23.1	19.2	13.4	10.6
Mean Daily Temperature Range	5% DB	MDBR	7.8	9.2	11.4	11.8	12.6	13.4	13.7	14.2	13.8	11.7	8.2	7.7
		MCDBR	9.8	13.7	17.5	18.0	18.7	19.6	19.5	20.3	18.8	15.7	11.1	9.2
		MCWBR	6.8	8.7	9.9	10.2	9.8	9.2	8.7	8.4	8.7	8.2	7.3	6.6
	5% WB	MCDBR	8.6	12.4	15.1	15.9	17.2	17.6	16.5	17.9	16.8	13.2	9.9	8.3
		MCWBR	6.3	8.1	8.8	9.1	9.3	8.5	7.9	7.9	8.0	7.3	6.8	6.2
Clear-Sky Solar Irradiance	taub	0.302	0.316	0.349	0.369	0.382	0.392	0.383	0.383	0.364	0.340	0.316	0.299	
	taud	2.528	2.480	2.398	2.349	2.339	2.341	2.364	2.373	2.427	2.500	2.526	2.539	
	Ebn at Noon	845	888	893	897	891	880	883	873	866	847	818	815	
	Edh at Noon	70	85	105	119	123	123	119	115	101	83	69	63	
All-Sky Solar Radiation	RadAvg	1.51	2.31	3.54	4.62	5.54	6.25	6.44	5.63	4.42	2.88	1.71	1.39	
	RadStd	0.17	0.31	0.46	0.48	0.43	0.39	0.31	0.30	0.21	0.26	0.19	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page